

Rocket Elevators

ELEVATE SAFETY, SERVICE AND VALUE

INTERNAL DOCUMENT

Inspection Types and What They Check

Internal document. Last updated April 2026. Contact: Compliance and Regulatory Affairs, Rocket Elevators Ontario Division.

This document covers every inspection type we encounter across our fleet of approximately 40,000 elevating devices in Ontario: what each inspection checks, how to prepare, and what the outcomes mean for our operations.

All inspections are performed by inspectors from the **Technical Standards and Safety Authority (TSSA)** under **Ontario Regulation 209/01**, evaluated against the **CSA B44 Safety Code for Elevators and Escalators**. Rocket Elevators does not perform inspections. Our role is to coordinate access, ensure a qualified technician is present, and act on the outcomes.

How Inspections Work at Rocket Elevators

When the TSSA schedules an inspection, the process follows a predictable cycle:

1. **Notification.** We receive notice from the TSSA. For periodic inspections, typically 2 to 4 weeks advance notice. For enforcement or incident inspections, notice may be shorter or nonexistent.
2. **Coordination.** We confirm the date with the building owner or property manager and assign one of our 14 field technicians to be on-site, ideally the one who regularly services the device.
3. **Preparation.** The technician prepares the device: runs the elevator to confirm basic operation, verifies the machine room and pit condition, and gathers maintenance records. See the preparation checklist below.
4. **Inspection.** The TSSA inspector conducts the inspection. Our technician operates the elevator as directed (running to specific floors, activating fire service, providing car-top access) and answers questions about maintenance history.
5. **Outcome.** If the device passes, we update records. If it fails, we enter the remediation cycle: assigning work to the appropriate contractor, tracking deadlines, and preparing for follow-up. See the outcomes section for details.

Our Contractors

We work with three primary contractors: **ThyssenKrupp Elevator** (traction and MRL elevators, our largest by volume), **KONE** (hydraulic fleet and modernizations), and **Schindler** (specialty installations including observation elevators). When an inspection produces compliance orders requiring contractor work, the zone manager assigns based on device type.

Preparing for Any Inspection

Regardless of the inspection type, the following preparation checklist applies. Technicians should complete these items before the inspector arrives. Many inspection failures are preventable with thorough preparation.

General Preparation Checklist

1. **Confirm scheduling.** Contact the building owner or property manager to confirm the date and time. Ensure they will provide access to the machine room, pit, and all floors served by the device. For multi-tenant buildings, coordinate with security or facilities management.
2. **Machine room readiness.** Clean and free of stored materials (TSSA inspectors cite storage frequently). Adequate lighting. Ventilation and temperature within range (10°C to 40°C). Main disconnect switch accessible and labeled. Oil levels checked. Door closes and locks properly.
3. **Pit readiness.** Clean, dry, free of debris. Sump pump operational (run it to confirm). Pit lighting working and switch accessible from pit entrance. Pit stop switch functional and labeled. Pit ladder in good condition and secured.
4. **Maintenance records.** Device log book available and up to date. Recent service reports from the past 12 months. Most recent inspection report. Documentation of any remediation work on outstanding compliance orders.
5. **Technician assignment.** Assign a technician who can operate the elevator in all modes (normal, inspection, fire service). Technician should arrive at least 30 minutes before the scheduled time.
6. **Keys and access.** Verify all required keys: machine room, car-top access, Phase II fire service, hoistway access (if applicable). Confirm the building has not changed locks without providing updated keys.
7. **Pre-inspection operational check.**
8. Run the elevator through its full travel. Listen for unusual noises, vibrations, or rough leveling.
9. Test the emergency phone or communication device. Confirm the connection is active and location information is correct.
10. Check door operation at multiple floors: opening, closing, reopening devices, nudging mode.
11. Test Phase I fire service recall and Phase II in-car operation.
12. Test emergency lighting (simulate power interruption or check battery indicator).
13. **Known issues.** Document any known issues and their remediation status. Have purchase orders or tracking information for parts on order. If a known issue affects safety, assess whether the device should be taken out of service before the inspection rather than risking a TSSA shutdown.

Periodic Inspection (ED-Periodic Inspection)

The **periodic inspection** is the most common inspection type in our system. Every active elevating device receives periodic inspections on a cycle determined by the TSSA. The frequency is based on the device type, age, usage, and risk profile. Most passenger elevators are inspected annually, though some high-risk or older devices may be inspected more frequently, and some low-risk devices may have extended intervals.

Periodic inspections are comprehensive. The inspector examines essentially every safety-relevant component, so any deferred maintenance or deteriorating condition is likely to be found.

What the Inspector Checks

The periodic inspection covers the device from machine room to pit.

Machine Room

- Cleanliness and organization. No storage of materials unrelated to the elevator.
- Temperature and ventilation within acceptable range (10°C to 40°C); ventilation system functional.
- Lighting adequate and functional.
- Main disconnect switch accessible, properly labeled, and in good working order.
- Controller condition: wiring integrity, relay operation (for older relay-logic controllers), fault log review (for microprocessor controllers).
- Motor and drive system: temperature, vibration, noise, current draw under load.
- Brake system: gap measurement between pads and drum, holding test (must hold car with full rated load), operational test (brake applies when power is removed).
- Oil levels on hydraulic systems, gear lubricant on geared traction machines.

Ropes, Chains, and Sheaves

- Wire rope inspection: broken wires (counted per rope lay), reduction in diameter, corrosion, lubrication. Ropes exceeding the allowable number of broken wires per lay must be replaced.
- Rope equalization: all ropes should carry approximately equal tension.
- Sheave condition: grooves must not be excessively worn. Worn sheaves accelerate rope deterioration.
- Compensation ropes or chains (on taller installations): condition and tension.
- For hydraulic elevators: cylinder condition, piping, and hose inspection.

Safety Devices

- **Governor:** tested by running the car at overspeed. Must trip and engage the car safety device.
- **Car safety device:** must grip the guide rails and stop the car. Tested with the governor.
- **Buffers:** oil buffers checked for oil level and operation; spring buffers checked for compression and return.
- **Overspeed protection:** verified for both up and down directions as required.

Hoistway

- Clearances between car and hoistway walls, car and counterweight, counterweight and walls. Must meet CSA B44 minimums.
- Guide rails: alignment, bracket condition, lubrication (where applicable).
- Counterweight: frame condition, filler weight security.
- Traveling cables: condition, routing, support, strain relief.

- Hoistway door interlocks at each landing (see Doors section below).

Pit

- Cleanliness, dryness, absence of debris.
- Buffer condition and positioning.
- Pit stop switch: accessible, clearly labeled, functional.
- Pit lighting and light switch.
- Sump pump operational (if equipped).
- Clearance below car and counterweight at lowest landing (CSA B44 minimum refuge spaces).

Car Interior and Car Top

- Car interior: condition of walls, floor, ceiling. Operating panel buttons readable and illuminated.
- Emergency lighting: must activate automatically when normal power is lost.
- Emergency communication device (phone or intercom): must connect to a monitored station.
- Ventilation fan operational.
- Car top: guardrails secure, stop switch functional and labeled, inspection station operational (run/stop, up/down, inspection speed), lighting adequate.

Doors

Door systems are one of the most frequently cited areas in inspection failures:

- **Interlocks:** every hoistway door interlock is tested. The car must not run unless all hoistway doors are closed and locked. This is the single most critical safety device on the elevator.
- **Closing force:** must not exceed the CSA B44 maximum (typically 135 N). Doors that close too forcefully are a hazard.
- **Reopening device:** door must reopen when an obstruction is detected.
- **Door restrictors (where equipped):** prevent the hoistway door from opening beyond a specified distance when the car is not at that floor.
- **Gap tolerances:** clearance between car door and hoistway door must be within limits.
- **Fire rating:** door assemblies must meet the required fire rating.
- **Nudging mode:** audible and visible signals when doors are held open; doors close at reduced speed and force.

Ride Quality and Performance

- **Leveling accuracy:** car must stop level with the landing sill within CSA B44 tolerance (typically ± 6 mm for passenger elevators). Poor leveling is a tripping hazard.
- **Speed:** must travel at rated speed within tolerance.
- **Acceleration and deceleration:** must be smooth and within comfort parameters.

Fire Service

- **Phase I (Firefighters' Emergency Operation):** the elevator must recall to the designated landing when activated, open its doors, and remain with doors open. All cars in a group must recall.
- **Phase II (Firefighters' In-Car Operation):** the elevator operates under manual control only. Doors do not open automatically; the car responds only to continuous pressure on a floor button.

Accessibility

- Floor designations on both jambs of each hoistway door: raised (tactile) and in Braille.
- Audible floor announcements (where required).
- Car operating panel button placement and labeling per accessibility standards.
- Emergency communication usable by persons with hearing disabilities (TTY or equivalent).

Duration and Scheduling

A typical periodic inspection takes 2 to 4 hours per device. Older devices with relay-logic controllers take longer than modern microprocessor-controlled installations. For buildings with multiple elevators, the TSSA typically inspects all devices during the same visit.

Initial Inspection (ED-Initial Inspection)

An **initial inspection** is performed before a newly installed device enters service. No device may carry passengers or freight until it has passed its initial inspection and received its TSSA license.

Who Arranges It

The installing contractor requests the initial inspection through the TSSA. Our role is oversight: we ensure the contractor requested the inspection, track scheduling, and have a technician available.

What the Inspector Checks

The initial inspection covers everything in a periodic inspection, plus installation-specific items:

- **Full code compliance.** Every component must meet the current edition of CSA B44 in effect at the time the permit was issued. No grandfathering for new installations.
- **Load testing.** Device tested at 150% of rated capacity to verify the car safety device, brakes, and structural components can handle overload. The contractor provides test weights.
- **Speed testing.** Car run at rated speed; governor overspeed switch trip speed verified.
- **Safety device testing.** Governor trip, car safety engagement, buffer, and all emergency stopping devices tested under controlled conditions.
- **Electrical testing.** Ground fault protection, circuit integrity, proper grounding.
- **Documentation review.** Installation drawings, engineering specifications, manufacturer certifications, wiring diagrams, and seismic risk assessment (if applicable).

Common Failure Reasons

Initial inspections have a higher failure rate than periodic inspections. Common reasons: incomplete punch list items, missing or non-functional safety devices (governor not calibrated, car safety not tested), electrical issues (improper grounding, wiring not matching drawings), documentation gaps (missing certifications, incomplete as-built drawings), door interlock adjustment issues, and construction debris remaining in the machine room or pit.

When a new installation fails, the installing contractor corrects all deficiencies before re-inspection. The device must not enter service until it passes.

Follow-up Inspection (ED-Followup Inspection)

A **follow-up inspection** occurs after a previous inspection identified violations and issued compliance orders. Its purpose is to verify that ordered corrections have been completed.

Scope

The follow-up is not a full re-inspection. The inspector checks only the items on the compliance orders (though new safety hazards noticed during the visit can be cited).

Possible Outcomes

- **All Orders Resolved / Passed:** every compliance order addressed. Device returns to normal status.
- **Follow up:** some orders resolved, others remain or remediation was inadequate. Another follow-up scheduled.
- **Fail:** orders not addressed or new significant issues discovered.

Our Timeline

Compliance orders come with deadlines. We must review orders immediately, assign the work to the appropriate contractor, track remediation against the deadline, and have documentation of completed work ready for the follow-up (contractor invoices, photos, test reports, updated maintenance records).

The follow-up is typically scheduled to coincide with the compliance deadline. If we cannot complete the work in time, we must request a time extension through the TSSA before the deadline passes, not after. See the operations handbook for compliance order handling and timeline management.

Related Follow-up Types

Several specialized follow-up types exist: **ED-Followup Lic Insp** (licensing requirements), **ED-Followup No-Lic Insp** (device found operating without a license), **ED-Followup Ownership Change** (building changed hands), **ED-Followup Minor Alt** (minor alteration), **ED-Followup Reg Non-Compliance** (regulatory non-compliance), **ED-FU Enforcement Action Insp** (enforcement action follow-up), and **ED-MCP Follow up** (maintenance control program). Each follows the same pattern: the inspector returns to verify previously identified issues have been resolved.

Major Alteration Inspection (ED-Major Alteration Inspection)

A **major alteration** is a significant modification affecting safety systems, structural integrity, or rated performance. Major alterations require a TSSA permit before work begins and an inspection before the device returns to service.

What Qualifies as a Major Alteration

Under Ontario Regulation 209/01, a major alteration includes replacement or modification of: the traction machine, motor, or hydraulic power unit; the controller (full replacement, not component-level repairs); wire ropes or hydraulic cylinder; car or counterweight safety devices; the governor; structural components (car frame, platform, guide rails, brackets); changes to rated capacity, speed, or travel distance; addition of floors served; or drive type conversion (e.g., hydraulic to traction).

Major alterations are most commonly triggered by modernization projects. When KONE modernizes an older hydraulic device or ThyssenKrupp replaces a controller and motor, the scope almost always qualifies.

Inspection Scope

Similar to an initial inspection, but focused on the altered systems and their integration with existing components. The inspector verifies that altered components meet the current CSA B44 (no grandfathering), integrate properly with unaltered systems, and pass load/speed testing if the alteration affected drive, motor, capacity, or speed. All documentation is required: alteration permit, engineering drawings, manufacturer certifications.

Our Role

The alteration contractor arranges the inspection through the TSSA. We verify the contractor obtained the permit before work began, track the timeline, ensure the device does not return to service before passing the inspection, and update our records with altered specifications.

A common mistake is allowing a device to return to service after the contractor declares the work "complete" but before the TSSA inspection. This is a regulatory violation.

Minor A and Minor B Inspections

Not every modification qualifies as a major alteration. Smaller changes are classified as either **Minor A** or **Minor B** depending on whether they involve safety-critical components.

ED-Minor A Inspection

A Minor A alteration involves safety-critical components that do not rise to the level of a major alteration: door operator replacement, safety device adjustments, interlock replacement, control system component replacement (e.g., a drive board, but not the entire controller), or emergency brake modifications.

Minor A alterations always require a TSSA inspection before the device returns to service. The inspector checks only the altered components and their integration with existing systems.

ED-Minor B Inspection

A Minor B alteration involves non-critical components: operating panel buttons, position indicators, cab interior renovation, ventilation modifications, or intercom/emergency phone replacement.

Minor B alterations may or may not require a TSSA inspection, depending on scope. When required, the inspection is limited to the altered components. Many Minor B alterations are handled by our technicians directly.

Practical Note

The line between Minor A and Minor B is not always obvious. When in doubt, treat the alteration as Minor A and request the inspection. It is far better to schedule an unnecessary inspection than to skip a required one and face enforcement action.

Enforcement Action (ED-Enforcement Action)

An **enforcement action** inspection is an unannounced visit by a TSSA inspector, triggered by compliance concerns. These carry significant operational and reputational implications.

What Triggers an Enforcement Action

Enforcement actions may be triggered by: complaints from building occupants or the public, a pattern of failed periodic inspections, intelligence from other regulatory sources, findings during inspections at the same property, reports of unlicensed operation, or reports of devices in service that should be shut down.

How They Work

We receive no advance notice. A TSSA inspector arrives at the building and requires immediate access. The building owner is legally obligated to provide access. If our technician is not on-site (and they usually are not, since these are unannounced), the inspector proceeds without one. The inspector has the authority to issue immediate compliance orders or shut down the device on the spot.

What to Do If an Enforcement Inspector Arrives

If a building owner or property manager contacts us to report an enforcement inspector:

1. Dispatch the nearest available technician immediately.
2. Instruct the building contact to provide full access. Do not delay or obstruct.
3. Notify the zone manager and Compliance and Regulatory Affairs.
4. If the technician arrives during the inspection, introduce themselves and cooperate fully.
5. Document the visit: inspector name, what was examined, orders issued, verbal instructions.

Why Enforcement Actions Matter

A pattern of enforcement actions suggests systemic failures in our maintenance program. Compliance and Regulatory Affairs reviews all enforcement actions quarterly. Zones showing a pattern require the zone manager to present a corrective action plan.

The **ED-FU Enforcement Action Insp** is the follow-up to verify that enforcement action issues have been resolved. These follow-ups are tracked with the same priority as the original enforcement action.

Incident and Near-Miss Inspections

When an incident or near-miss is reported to the TSSA, they may dispatch an inspector to examine the device and investigate.

ED-Perform L1 Incident Insp

A **Level 1 incident inspection** is triggered by a reported incident involving injury, entrapment, or significant equipment failure. The inspector:

- Examines the device in its current state (this is why preserving the scene is critical)
- Reviews maintenance records and the recent inspection history
- Interviews witnesses if present
- Determines the probable root cause of the incident
- May order the device shut down pending further investigation or repairs

ED-Perform L1 Near Miss Insp

A **Level 1 near-miss inspection** follows a reported event that could have resulted in injury but did not. The same investigation process applies, though the urgency may be lower since no injury occurred. Near-miss inspections are important because they often identify conditions that, if left unaddressed, would lead to an actual incident.

Our Role During Incident and Near-Miss Inspections

1. **Preserve the scene.** Do not reset, repair, or adjust anything until the TSSA inspector has examined it. If passengers are trapped, follow the entrapment rescue procedure first, then preserve the scene.
2. **Provide maintenance records.** The inspector will want the device log book, recent service reports, the last inspection report, and any open compliance orders.
3. **Cooperate fully.** Answer all questions honestly. Do not speculate about causes; report what is known.
4. **Document everything.** Record the timeline of events, who was notified, actions taken, and the inspector's findings.

These inspections feed into the TSSA's root cause classification system, which categorizes incidents by contributing factors. Our internal data mirrors these categories (component failure, maintenance-related, environmental, installation error, vandalism, or unknown) for our own risk analysis.

Re-Activation Inspection (ED-Re-Activate Inspection)

A **re-activation inspection** is required before any shut-down device can return to service, whether the shutdown was TSSA-initiated or voluntary (sometimes called a **Customer Shutdown**).

Scope

The inspection scope depends on how long the device has been out of service:

- **Under 6 months:** inspector focuses on the shutdown reason and verifies it has been resolved, plus a general safety assessment.

- **6 months to 1 year:** expands to cover most of a periodic inspection, since idle components may have deteriorated.
- **Over 1 year:** essentially an initial inspection. Ropes may have corroded, hydraulic seals may have dried out, electrical connections may have corroded, and control systems may have developed faults.

The Re-Activation Process

1. Complete all remediation work that prompted the shutdown (or maintenance required to bring the device to operating condition).
2. Submit a re-activation request to the TSSA.
3. The TSSA schedules the re-activation inspection.
4. Prepare as for a periodic inspection, with attention to items that may have deteriorated during the shutdown.
5. Pass the inspection. The device receives clearance to return to service.

The device must not carry passengers or freight until the TSSA has issued clearance. This is non-negotiable, regardless of pressure from the building owner.

Other Inspection Types

The following inspection types appear less frequently in our records but are important to understand when they occur.

ED-Unscheduled Inspection

A TSSA-initiated inspection outside the regular periodic cycle. Unlike enforcement actions, these are not necessarily triggered by a compliance concern. The TSSA may conduct them as part of targeted campaigns (focusing on a device type or age range), random audits, or in response to industry-wide safety bulletins.

ED-Sub Inspection and ED-Sub Inspection Major

Performed by inspectors subcontracted by the TSSA to handle volume during peak periods. Same standards and procedures as TSSA staff inspectors. From our perspective, preparation and response are identical.

ED-PWGSC Insp and ED-PWGSC Foll-Up

Inspections of elevating devices on federal government properties. **PWGSC** (Public Works and Government Services Canada) properties are inspected by the TSSA under a federal service agreement. These follow CSA B44 but may include additional federal requirements. The follow-up variant verifies resolution of issues from the initial inspection.

ED-Non-Mandated Insp ON and ED-Non-Mandated Followup ON

Voluntary inspections requested by a building owner or by us. Useful for independent assessment before a modernization decision or after significant repair work. The follow-up variant tracks resolution of any issues found.

ED-Inspection Temp Lic

Inspection for a **temporary license**, issued when a device must operate for a limited period under specific conditions (phased construction, waiting for parts). Verifies the device is safe under the temporary conditions.

ED-Followup Ownership Change

After a building changes hands, the TSSA may require an inspection to verify the new owner is maintaining the device to standard. Common when the ownership change involves a change in elevator service provider.

ED-Reg Non-Compliance and ED-Followup Reg Non-Compliance

Initiated when the TSSA identifies a potential violation of Ontario Regulation 209/01 beyond device-specific code issues: operating without a license, failure to report incidents, failure to maintain records. The follow-up verifies correction.

ED-MCP Enforcement Insp and ED-MCP Follow up

MCP (Maintenance Control Program) inspections verify that a device enrolled in an approved MCP under Ontario Regulation 209/01 is being maintained according to the program. Non-compliance may result in enforcement action.

Understanding Inspection Outcomes

Every inspection produces an outcome that determines what happens next. Below is a practical guide to each outcome and the action we take.

Passed

The device meets all applicable requirements. No compliance orders, no deficiencies noted.

Our action: Update the device record in our system with the inspection date and result. No further action required until the next periodic inspection. Confirm with the building owner that the device passed.

Fail

The inspector found violations and issued **compliance orders**. Each order identifies the specific violation, the code reference, and the deadline for correction.

Our action: Review all orders immediately and categorize by severity and deadline. Assign remediation work to the appropriate contractor (ThyssenKrupp, KONE, or Schindler based on device type). Enter each order into our compliance tracking system. Track contractor progress and prepare documentation for the follow-up inspection. See the operations handbook for compliance order handling and risk scoring.

Follow up

Some compliance orders from the previous inspection were resolved, but others remain outstanding, or the inspector found that some corrections were inadequate. A subsequent follow-up inspection will be scheduled.

Our action: Same as Fail, focused on the remaining open orders. Review what was accepted and what was not. If a correction was rejected, determine why and redo the work. Track the next follow-up deadline.

Shutdown

The inspector found **severe violations** presenting an immediate safety hazard. The device is ordered out of service immediately. Can occur during any inspection type.

Our action: This triggers the TSSA Shutdown Response procedure (see Emergency Response Protocols). Device is taken out of service immediately with "Out of Service" signage posted. Notify the building owner, zone manager, and Compliance and Regulatory Affairs. Develop a remediation plan, complete the work, submit a re-activation request, and pass the re-activation inspection before the device returns to service. Shutdowns are tracked as critical compliance events and reviewed by senior management.

Unable to Inspect

The inspector could not complete the inspection. Common reasons: access denied, device not operational (already shut down or under construction), unsafe conditions, or no qualified person available to operate the elevator.

Our action: Reschedule as soon as possible. Determine and correct the underlying cause (access coordination failure, technician scheduling gap). Repeated "Unable to Inspect" results can trigger enforcement action, which is a significantly worse outcome.

All Orders Resolved

All outstanding compliance orders from previous inspections have been satisfactorily addressed and closed. The device returns to a clean compliance status.

Our action: Update the device record. Close all open compliance orders in our tracking system. Confirm with the building owner.

Administrative Outcomes

Several outcomes in the TSSA system are administrative rather than inspection results. These include: **Cancelled** (scheduled but cancelled, may need rescheduling), **Dismantled** (device removed, update records), **DC Follow up** (Director's Compliance order follow-up), **Extend Time to Comply** (TSSA granted a deadline extension), **Order Transferred** (orders moved to a different responsible party, e.g., after ownership change), **Received** (TSSA received documentation, awaiting processing), **RC Established** (Responsible Contact registered for the device), **Temp Lic Not Needed** (temporary license application deemed unnecessary), and **Complete / Inspection Complete** (inspection process finished, final result recorded separately).

Common Reasons for Failed Inspections

Based on our fleet experience, these are the most frequent reasons for failed periodic inspections.

Door Interlock Issues

Door interlocks are the most frequently cited violation across our fleet. Common problems include worn contacts that no longer make reliable connection, misadjusted rollers that allow the door to be opened slightly

with the car away from the landing, and bent or damaged assemblies from impact or vandalism. Door interlocks should be checked during every routine maintenance visit, not just before inspections.

Pit Conditions

Pit-related violations are the second most common category: water accumulation (sump pump failure, drainage issues, groundwater), debris (construction materials, garbage, rodent activity), non-functional pit stop switches, inadequate lighting, and unsecured pit ladders. Building owners sometimes use elevator pits as unofficial storage. Our technicians should flag pit conditions during every visit and report them to the building owner in writing.

Machine Room Issues

Machine rooms are cited for storing materials unrelated to the elevator, inadequate ventilation or excessive temperature, uncleaned oil leaks, insufficient lighting, and disconnect switches blocked by stored materials.

Emergency Communication Failures

The emergency phone or intercom must connect to a monitored station at all times. Common failures: phone line disconnected (building telecom changes, unpaid monitoring service), handset damaged or missing, incorrect location information at the monitoring station, and non-functional TTY or accessibility communication.

Fire Service Test Records

Common violations: no record of Phase I and Phase II testing in the log book, expired testing records, and incomplete records (only one phase tested). Fire service testing should be performed and documented during routine maintenance visits per the maintenance contract schedule.

Rope Wear

Wire rope deterioration beyond allowable limits triggers a fail and often leads to mandatory replacement. Indicators: broken wires exceeding the allowable count per rope lay, reduction in diameter indicating internal wear, corrosion (particularly in humid environments), and uneven wear patterns suggesting sheave or alignment problems. Ropes approaching their limits should be flagged for proactive replacement.

Emergency Lighting

Emergency lighting must activate automatically when normal power is lost. Common failures: dead or weak battery packs, burned-out bulbs, and non-functional transfer circuitry (goes unnoticed because normal power is present). Test during every routine maintenance visit by briefly disconnecting the car lighting circuit.

Device Type Considerations

Our fleet includes several device types, each affecting inspection scope and preparation:

- **Passenger Elevators** (~72% of our fleet): standard device type, follows the periodic inspection process described above.
- **Freight Elevators** (E-type and P-type): additional requirements for rated load, door configuration (often vertical bi-parting), and industrial environment hazards (dust, chemical exposure, vibration).

- **Observation Elevators:** glass panels add inspection points for glass integrity, mounting security, and hoistway visibility. Schindler handles most of these installations.
- **LULA Elevators** (Limited Use/Limited Application): smaller, lower-speed devices for small buildings or accessibility. Own subset of CSA B44 requirements; shorter inspections.
- **Sidewalk Elevators:** vertical access between sidewalk and basement. Unique weather exposure and security considerations.
- **Material Lifts (ATD), Power Type Manlifts, and Special Installations:** specialized CSA B44 sections apply. Technicians must be familiar with the specific requirements.
- **Temporary Elevators:** used during construction, licensed under temporary licenses, inspected as ED-Inspection Temp Lic.

Quick Reference: Inspection Types at a Glance

INSPECTION TYPE	TRIGGER	NOTICE	SCOPE	TYPICAL DURATION
ED-Periodic Inspection	Regular cycle	2-4 weeks	Full device	2-4 hours
ED-Initial Inspection	New installation	Contractor arranged	Full device + load/speed tests	4-6 hours
ED-Followup Inspection	Previous failure	1-2 weeks	Compliance orders only	1-2 hours
ED-Major Alteration Inspection	Major modification	Contractor arranged	Altered systems + integration	2-4 hours
ED-Minor A Inspection	Safety-critical modification	Contractor arranged	Altered components	1-2 hours
ED-Minor B Inspection	Non-critical modification	As scheduled	Altered components	30 min - 1 hour
ED-Enforcement Action	Compliance concerns	None (unannounced)	As determined by inspector	Varies
ED-Perform L1 Incident Insp	Reported incident	Immediate	Incident-related components	Varies
ED-Perform L1 Near Miss Insp	Reported near-miss	Short notice	Event-related components	Varies
ED-Re-Activate Inspection	Return from shutdown	After request submitted	Depends on shutdown duration	2-6 hours
ED-Unscheduled Inspection	TSSA initiative	Varies	As determined by inspector	Varies

For specific CSA B44 code requirements referenced during inspections, see the Safety Code Quick Reference. For compliance order handling workflows and risk scoring, see the operations handbook. For procedures following a TSSA shutdown order, see Emergency Response Protocols.